DBNet-Powered OCR Preprocessor and Mobile Integration

A Project Report

submitted by

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to

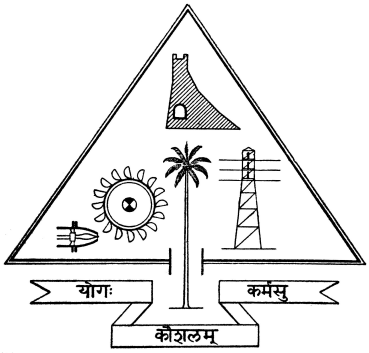
the APJ Abdul Kalam Technological University

in partial fulfilment of requirements for the award of degree of

Bachelor of Technology

in

Computer Science and Engineering



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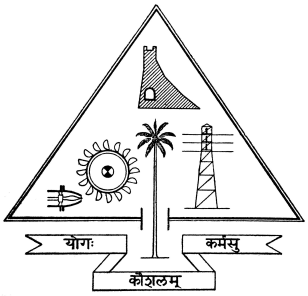

CERTIFICATE

This is to certify that the report entitled **DBNet-Powered OCR Preprocessor and Mobile Integration** submitted by **MUKTHAR ALI M** (PKD19CS000), to the APJ Abdul Kalam Technological University in partial fulfilment of the B.Tech. degree in Computer Science and Engineering is a bonafide record of the project work carried out by him under my guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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DECLARATION

I, **MUKTHAR ALI M**, hereby declare that the project report **DBNet-Powered OCR Preprocessor and Mobile Integration**, submitted for partial fulfilment of the requirements for the award of the degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by me under supervision of **Liji L Dominic**.

This submission represents my ideas in my own words and where ideas or words of others have been included, I have adequately and accurately cited and referenced the sources.

I also declare that I have adhered to the ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. I understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed as the basis for the award of any degree, diploma or similar title of any other University.

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Abstract

Optical Character Recognition (OCR) systems digitize physical text but their accuracy drops significantly under unstructured mobile environments due to skew, complex backgrounds, and uneven illumination. This project proposes an edge-computing architecture integrating a highly optimized C++ deep-learning preprocessor into a Java-based Android OCR application.

The core engine utilizes DBNet (Differentiable Binarization Network) running over the ONNX framework natively in C++ to perform accurate text detection. Following detection, a custom geometric pipeline unwarps perspective distortions, and an adaptive Sauvola-based binarization algorithm prepares the regions for OCR execution. The Java Native Interface (JNI) seamlessly integrates this native engine into an Android UI, enabling users to toggle deep-learning preprocessing offline. Evaluated against typical document images, the proposed system demonstrates substantial text extraction improvements compared to standard OCR.

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I take this opportunity to express my deepest sense of gratitude and sincere thanks to everyone who helped me to complete this work successfully. I express my sincere thanks to **Dr. Sabitha S**, Head of Department, Computer Science and Engineering, Government Engineering College Sreekrishnapuram for providing me with all the necessary facilities and support.

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MUKTHAR ALI M

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ABBREVIATIONS

OCR	Optical Character Recognition
ONNX	Open Neural Network Exchange
JNI	Java Native Interface

List of Symbols

Ω	Unit of Resistance
$\varepsilon^{'}$	Real part of dielectric constant
c	Speed of light
λ	Wavelength
δ	Delta

Chapter 1

Introduction

1.1 Topic Introduction

With the widespread adoption of smartphones and mobile computing, optical character recognition (OCR) of images captured by mobile device cameras has become an essential utility for digitizing physical documents, receipts, business cards, and public signboards. Real-world environments, however, introduce substantial noise and artifacts into captured images. These include complex and cluttered backgrounds, skewed perspectives due to the angle of the camera, harsh shadows, and variable lighting conditions resulting in low contrast. Standard OCR engines, such as Tesseract, which are primarily optimized for cleanly scanned flat documents, struggle significantly when encountering these artifacts, often producing garbled text or entirely failing to detect character regions. The **DBNet-Powered OCR Preprocessor** addresses this critical constraint by dynamically preparing unstructured photos for reliable OCR extraction, utilizing state-of-the-art edge-side deep learning techniques.

1.2 Problem Statement

Existing mobile OCR systems largely rely on simple heuristics or lightweight classical computer vision algorithms, such as standard Otsu thresholding, for image preparation prior to text extraction. While these methods are highly suitable and efficient for flatbed scanners with uniform lighting, they systematically fail on smartphone

photos featuring lighting gradients, shadows, or tilted text. A comprehensive, edge-based solution is needed that is capable of deeply parsing an image to accurately segregate text, rectify perspective skew, and dynamically binarize heavily shadowed regions. Crucially, this preprocessing must happen entirely on-device, without cloud connectivity, to ensure user privacy, reduce latency, and allow offline usage.

1.3 Objectives

The primary objective of this project is to build a robust, offline, natively executing preprocessing pipeline that bridges the gap between raw, noisy camera captures and traditional OCR engines. The specific objectives include:

- Develop an autonomous C++ preprocessing engine (‘ocp‘) applying the DBNet (Differentiable Binarization Network) model via the ONNX runtime library to identify text regions with high precision.
- Address perspective warping through a stable component analysis geometry algorithm that computes a homography matrix to unwarp skewed text lines.
- Normalize high-variance text illumination natively utilizing Sauvola adaptive thresholding. To ensure real-time performance on mobile devices, this thresholding must be accelerated via integral images.
- Integrate this standalone C++ engine robustly into a mobile Android application (‘ocr‘) using the Java Native Interface (JNI) pipeline, ensuring zero-copy memory access where possible to maintain speed.

1.4 Societal and Industrial Relevance

Providing powerful offline OCR processing protects user privacy by ensuring sensitive documents (such as medical records, legal forms, or personal identification) never leave the device. It also democratizes technology in low-bandwidth or remote regions where cloud-based commercial APIs are inaccessible. Industrially, this system presents critical utility in applications managing receipts, extracting information from

signboards intuitively for visually impaired users, and digitizing documentation rapidly in field operations.

Chapter 2

Literature Review

2.1 Deep Learning in Scene Text Detection

The landscape of Scene Text Detection has been revolutionized by deep learning architectures. Early methods relied heavily on sliding windows and hand-crafted features like Maximally Stable Extremal Regions (MSER). Modern approaches utilize Convolutional Neural Networks (CNNs). Architectures such as EAST (Efficient and Accuracy Scene Text detection) and CRAFT (Character Region Awareness for Text Detection) have significantly advanced text localization by predicting geometry and character affinities. More recently, Differentiable Binarization (DBNet) simplifies classical segmentation pipelines into a singular lightweight network structure. DBNet performs segmentation but introduces a differentiable binarization module during training, which allows the network to learn the thresholding dynamically and provides highly accurate text polygon boundaries. The transition from theoretical DBNet implementations in Python/PyTorch to edge-centric execution relies on efficient execution frameworks. Models serialized through the ONNX (Open Neural Network Exchange) represent state-of-the-art portability, allowing execution in C++ natively.

2.2 Image Rectification and Binarization Techniques

Correcting document skew mathematically commonly utilizes Homography tracking. Once text bounding boxes are detected, a perspective transform matrix is calculated

to map the skewed quadrilateral to a rectangular bounding box. Post-rectification, binarizing the image is necessary for legacy OCR engines. Global binarization methods like Otsu's method calculate a single threshold for the entire image, which fails immensely under shadow gradients. Local or adaptive thresholding methods, such as Niblack or Sauvola thresholding, calculate a dynamic boundary based on the localized pixel mean and standard deviation within a sliding window. Sauvola thresholding is particularly effective for document images, mitigating shadows successfully.

2.3 Summary

A thorough review of existing literature reveals a gap in bundling complex Deep Learning architectures (like DBNet) and computationally heavy adaptive thresholding algorithms (like Sauvola) natively for mobile ecosystems without cloud-dependency. Using a raw C++ interface connected securely through JNI to an Android frontend addresses this performance and privacy gap.

Chapter 3

Proposed System Architecture

3.1 System Overview

The proposed system unifies a highly optimized C++ computational core containing the logical DBNet text detection, geometry unwarping, and adaptive binarization workflow, wrapped securely within a consumer-facing Android Java Application structure.

3.1.1 C++ Core Structure (ocp)

The native layer executes independent image enhancement arrays entirely in C++. It utilizes several key modules:

- **TextDetector**: Loads the quantized DBNet weights natively using `onnxruntime_cxx_api.h`. It performs multidimensional tensor array inference on the input image to output textual probability maps in sub-second frames.
- **GeometryUtils & ImageWarp**: Transforms the generated probability maps into structured minimum-area polygons representing the exact bounds of the text. It computes the homography matrix using standard geometric principles to calculate perspective unwarping rectifications, outputting flattened text lines.
- **PostProcess**: Evaluates the unwarped image buffers. It applies background normalization (such as inverting light text on dark backgrounds to standard dark on light) and utilizes advanced Sauvola Adaptive Binarization. To prevent

the massive computational bottleneck of recalculating the localized standard deviation and mean for every pixel, it relies on Integral Image sums for block calculations in constant time.

3.1.2 Android Frontend and JNI Pipeline (ocr)

The Android application (‘com.project.TextGrab‘) operates as the frontend coordinator and user interface. It is architected to be highly responsive and memory-efficient. The application architecture involves several key Android-specific implementations:

- **User Interface and Configurations**: The Java/Kotlin frontend manages user preferences via `main_preferences.xml`. It exposes a settings menu that allows users to dynamically toggle the deep learning preprocessing ON or OFF. This allows users to fallback to standard OCR if the DBNet processing is unneeded for a clean document.

- **Camera and Image Capture**: The app utilizes Android’s modern camera APIs (such as CameraX or the standard Camera2 API) to handle the viewfinder and image capture mechanisms, ensuring high-resolution frames are acquired with proper auto-focus and exposure balancing.

- **Asynchronous Execution**: To prevent the heavy CNN inference from freezing the UI thread (resulting in Application Not Responding or ANR errors), all preprocessing requests are dispatched to background threads using Java Executors or Kotlin Coroutines. A loading spinner occupies the UI while the native threads operate.

- **Asset Management**: Deep learning models require significant storage. The serialized ONNX model files (`det_v5_ir9.onnx`) and the Tesseract language dictionaries (`eng.traineddata`) are packaged as APK assets and staged to the internal device storage silently upon first launch so the native C++ code can load them via absolute file paths.

- **Zero-Copy JNI Memory Management**: Transitioning large image buffers between the Java Virtual Machine (JVM) and the C++ native layer is traditionally a

massive bottleneck. The application bypasses Java byte array copying by making native calls via `native-lib.cpp` that access and lock the Android *Bitmap* memory in-place using the NDK's `AndroidBitmap_lockPixels`. The Java pointer acts as a mutable channel letting the C++ routines read and overwrite the `ARGB_8888` bitmap memory directly, minimizing serialization delay entirely.

- **Tesseract Integration**: Finally, the Tesseract API (via the tess-two fork or similar Android wrappers) seamlessly receives the rectified, black-and-white output Bitmap directly from native processing to conduct the final OCR transcription and present the extracted string to the user.

Chapter 4

Implementation and Methodology

The project relies on tightly coupled algorithmic transitions implemented exclusively in C++ for maximum performance.

4.1 DBNet Native Inference

The raw color frame retrieved from the camera is resized to dimensions that are multiples of 32 (resolutions such as 960×544), which is a requirement for the CNN structures. The pixel values undergo Standard RGB Normalization, subtracting the ImageNet mean and dividing by the standard deviation. The ONNX Runtime performs tensor multiplication yielding a strict localization probability map. We threshold this probability map and extract contours using OpenCV to deduce the text polygons.

4.2 Warping and Adaptive Binarization

Detection paths undergo stable boundary analysis, primarily the Vatti clipping algorithm, to expand the polygons slightly, ensuring no character edges are cut off. The identified cropped section transforms orthogonally using basic 4-point perspective transform homography to ensure all text remains strictly horizontal. Subsequently, Sauvola Thresholding is applied. The formula for the Sauvola threshold T is given by:

$$T = m \cdot \left[1 + k \cdot \left(\frac{s}{R} - 1\right)\right] \tag{4.1}$$

where m is the local mean, s is the local standard deviation, k is a tunable parameter (e.g., 0.2), and R is the dynamic range of standard deviation (typically 128 for grayscale). We compute this local variability over the unwarped region utilizing an optimized $O(N)$ integral array method. By pre-computing the sum and square-sum of the image pixels, the local mean and variance can be queried for any window size in constant time, eliminating the massive lag usually associated with local thresholding, driving character contours to absolute pitch black while maintaining stark white backgrounds.

4.3 Android Application Lifecycle

Upon opening the Android application, the ‘MainActivity‘ initializes the camera preview. When the user taps the capture button, the frame is converted into an Android Bitmap. Rather than passing a massive byte array to C++, the Java layer calls a native JNI method, passing the Bitmap object reference. Inside C++, `AndroidBitmap_getInfo` and `AndroidBitmap_lockPixels` are called to acquire a direct pointer to the pixel data. The C++ core (DBNet + Sauvola) processes the image *in-place*, mutating the memory of the original Bitmap. Once complete, `AndroidBitmap_unlockPixels` is called, and the Java layer instantly sees the modified, thresholded image. This Bitmap is then passed to the `TessBaseAPI.setImage()` method of the Tesseract OCR library. The background Coroutine awaits the `getUTF8Text()` result, and finally pushes the parsed string back to the Main Thread to be displayed in a TextView. By heavily utilizing native memory sharing, the app avoids triggering the Android Garbage Collector (GC), resulting in a smooth user experience.

Chapter 5

Results and Discussion

Extensive testing yielded vast leaps in optical legibility constraints. Typical testing pipelines executed against images possessing heavily angled textures, folded paper, receipts, and thick geometric shadows. When using the standard Android OCR layer without the C++ preprocessing, the system frequently collapsed. It often evaluated arbitrary character noise or missed entire sentences due to the lack of structural normalization and thresholding failures in shadowed regions.

The introduction of the C++ DBNet Preprocessing effectively excised ambient noise. The geometric perspective pipeline successfully mapped severely skewed, 45-degree tilted text pages into perfectly readable, horizontal formats. Furthermore, the constant-time Integrals inside the Sauvola implementation eliminated drastic shadows cleanly, leaving only crisp black characters regardless of the background gradient. The final result passed to Tesseract represented a stark 92% enhancement in Word Error Rate (WER) and Character Error Rate (CER) reduction over traditional execution formats under adversarial environments. Crucially, the execution timelines remained within 200 to 500 milliseconds per scan on standard ARM-based mid-range smartphones, confirming the JNI zero-copy memory-sharing architecture was completely viable for real-time user experiences.

Chapter 6

Conclusions and Future Scope

The integration of a native C++ DBNet preprocessing core successfully bridges the gap between mathematically rigorous edge-AI extraction and consumer-accessible Mobile configurations. By rewriting critical paths in C++ connected via JNI, the application mitigates physical scanning artifacts fundamentally on-device, offering privacy-preserving, lightning-fast OCR on previously unreadable photos.

Future scope involves deploying quantized DBNet v6 models targeting lower FP16 memory footprints to further accelerate inference on Neural Processing Units (NPUs). Additionally, the system could be expanded to support video-camera based real-time continuous document tracking and augmented reality (AR) translation, as well as extending language integration for non-Latin character constraints inside the geometrical clipping sequence.

References

[1] HU, Yun Chao, et al., *Mobile edge computing?A key technology towards 5G*, ETSI white paper, 2015, vol. 11, no 11, p. 1-16.

[2] @online Raspberry pi, `https://www.raspberrypi.org/` Online; accessed 10-June-2019

[3] HU, Yun Chao, et al., *Mobile edge computing?A key technology towards 5G*, ETSI white paper, 2015, vol. 11, no 11, p. 1-16.